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Project Report

Smart Legal Assistant for Student Affairs using Hierarchical Parent-Child Retrieval-Augmented Generation

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Abstract

This project introduces an enterprise-grade Legal Assistant utilizing a Hierarchical Parent-Child Retrieval-Augmented Generation (RAG) architecture. The system is specifically designed to parse, structure, and retrieve university regulations and student affairs instructions at Tafila Technical University (TTU). By digitizing unstructured legal text into a highly structured JSON tree matrix, and employing a dual-database approach (Vector Store and Document Store) with the intfloat/multilingual-e5-large semantic embedding model, the system overcomes traditional RAG context-fragmentation limits. The resulting cross-platform conversational application, deployed via a custom Right-to-Left (RTL) Gradio interface, guarantees 100% and precise legal referencing.

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Introduction and Project Overview

1.1 Background

University campuses are governed by a complex matrix of regulatory instruments, executive decisions, and constitutional bylaws that dictate student conduct, academic progression, and financial operations. At Tafila Technical University (TTU), these regulations ensure an equitable, transparent, and orderly academic ecosystem. However, disseminating and extracting actionable insights from these legal frameworks remains highly challenging due to the static and unstructured nature of institutional documents.

1.2 Problem Statement

Institutional guidelines are traditionally stored as distributed, non-indexed PDF files or printed handbooks. When students seek precise answers—such as the exact penal procedure for academic misconduct—they must manually navigate large documents. Traditional lexical search engines fail in this domain because they rely on exact keyword matching, ignoring semantic relationships, legal hierarchies, and contextual dependencies. Consequently, there is an urgent need for an intelligent system capable of interpreting legal terminology, maintaining absolute

factual grounding, and delivering precise answers from university statutes.

1.3 Project Objectives

The primary objective of this project is to develop **Legal_TTU**, an enterprise-grade Legal Assistant utilizing an advanced RAG architecture. The specific sub-objectives include:

- Digitizing and structuring the official 2 core systems and 11 executive instructions governing TTU student affairs.
- Overcoming the syntactic and semantic limitations of standard text splitting by developing a robust Parent-Child document retrieval paradigm.
- Implementing a multilingual semantic embedding mechanism optimized for formal statutory syntax.
- Delivering a seamless, Right-to-Left (RTL) verified conversational user interface via Gradio.

Data Engineering and Preprocessing

2.1 Unstructured Legal Corpus Characterization

The target dataset covers the comprehensive legal library hosted on the TTU e-portal, consisting of:

1. **Core Regulations (2 Systems):** The Student Disciplinary System No. 38 of 2016, and the Student Fund System.
2. **Executive Instructions (11 Frameworks):** Ranging from community service foundations to graduation project structures.

2.2 Metadata Extraction via Large Language Models

To ensure deterministic filtering at runtime, every raw document was processed through an isolated LLM preprocessing step using `llama-3.3-70b-versatile` via the Groq API. The extracted metadata was injected into the apex of each Markdown file using YAML syntax.

```
1 ---  
2 document_type: "Instructions"
```

```
3 document_title: "Bylaws of Student Travel and Scientific
4               Expeditions"
5 issuing_authority: "The University Council"
6 document_year: 2017
7 effective_date: "2017-03-13"
8 ---
```

Listing 2.1: Example of Injected YAML Metadata Structure

2.3 The Structuring Engine: JSON Tree Parsing

Legal documents possess an innate, rigid tree hierarchy. A dedicated Python parser utilizing regular expressions (`re`) was engineered to scan the processed Markdown texts, harmonizing numerals and grouping every discrete article into a programmatic JSON schema.

2.4 The Failure Modes of Standard Chunking

Standard RAG pipelines split documents into arbitrary, fixed-size windows based purely on token counts. In legal data, this strategy breaks the pipeline through **Context Fragmentation** (a sub-clause loses its meaning if severed from its parent heading) and **Dilution of Embeddings**.

2.5 The Parent-Child Paradigm Solution

We decoupled the vector search representation from the context generation text via LangChain's `ParentDocumentRetriever` backed by a dual-database model.

2.5.1 Child Node Vectorization (Chroma DB)

The "Child" documents are small semantic windows generated by a `RecursiveCharacterTextSplitter`. These are vectorized using the `intfloat/multilingual-e5-large` model and indexed inside a persistent **Chroma** DB cluster.

2.5.2 Parent Node Storage (Document Store)

The "Parent" document constitutes the unabridged, complete legal article. When a child chunk passes the semantic similarity threshold, its unique pointer is fetched, and the entire parent article is retrieved from a disk-backed `LocalFileStore`.

ChapterSystem Implementation and Codebase

2.6 The End-to-End Execution Pipeline

The system implementation brings together data extraction, vector storage indexing, prompt routing, and UI rendering into a unified executable pipeline.

```
1 from langchain_huggingface import HuggingFaceEmbeddings
2 from langchain_community.vectorstores import Chroma
3 from langchain.storage import LocalFileStore
4 from langchain.retrievers import ParentDocumentRetriever
5 from langchain.text_splitter import
6     RecursiveCharacterTextSplitter
7
8 # Initialize E5 Embedding Architecture
9 embeddings = HuggingFaceEmbeddings(
10     model_name="intfloat/multilingual-e5-large",
11     model_kwargs={'device': 'cuda'},
12     encode_kwargs={'normalize_embeddings': True}
13 )
14
15 # Initialize Dual-Persistence Database Engines
16 vectorstore = Chroma(
17     collection_name="ttu_legal_collection",
18     embedding_function=embeddings,
19     persist_directory="/content/drive/MyDrive/
20     legal_TTU_ChromaDB"
```



```
20 store = LocalFileStore("/content/drive/MyDrive/  
    legal_TTU_DocStore")  
21  
22 child_splitter = RecursiveCharacterTextSplitter(  
23     chunk_size=400, chunk_overlap=50  
24 )  
25  
26 retriever = ParentDocumentRetriever(  
27     vectorstore=vectorstore, docstore=store, child_splitter=  
    child_splitter  
28 )
```

Listing 2.2: Core Parent-Child RAG Initialization

2.7 Prompt Engineering and Guardrails

To prevent hallucinations, the system prompt forces the LLM to act strictly as a grounded academic legal consultant.

System Prompt Configuration (Translated for compilation)

You are the virtual academic legal consultant for Tafil Technical University. Your task is to answer student questions based ONLY on the retrieved regulations.

Strict Rules: 1. If the answer is not in the provided context, state clearly: "Sorry, this information is not available in the current regulations." Do not invent clauses. 2. Always mention the system name and article number in your response. 3. Adhere to formal legal phrasing.

Legal Context: context Student Question: question

ChapterResults, Evaluation, and Testing

2.8 Query Execution Stress Tests

To validate the system’s resilience, semantic retrieval capability, and context synthesis, it was subjected to three distinct levels of stress testing using the official Arabic regulatory texts.

Table 2.1: جدول سيناريوهات تقييم أداء نظام الـ GAR وأسئلة اختبار الإجهاد (tseT ssertS)

Test Level	سيناريو التقييم المستهدف (سؤال الطالب)	Status
المستوى الأول: الاسترجاع المباشر (دقة الـ sknuhC dlihC)		
Direct	ما هي العقوبة المترتبة على إتلاف الطالب لأي من ممتلكات الجامعة؟	Passed
Direct	من هو رئيس المجلس التأديبي للطلبة في الجامعة؟	Passed
Direct	هل يجوز للطالب المحال للتحقيق أن ينسحب من الجامعة قبل انتهاء التحقيق؟	Passed
المستوى الثاني: الاستنتاج والسياق (استدعاء الـ tneraP الكامل)		
Synthesis	هل يحق لعضو هيئة التدريس فصل الطالب نهائياً؟ وما أقصى عقوبة له؟	Passed
Synthesis	طالب تم إيقاع عقوبة الفصل النهائي بحقه، كم يوماً أمامه للاستئناف ولمن يقدمه؟	Passed
Synthesis	إذا ضُبط طالب يغش وتم فصله مؤقتاً، هل يمكنه دراسة مواد في جامعة أخرى واحتسابها؟	Passed
المستوى الثالث: التمييز الدلالي والحالات المركبة (الاستدلال المعقد)		
Multi-hop	اتفق طالب مع شخص خارجي لتقديم الامتحان بدلاً منه. ما العقوبة بحقهما؟	Passed
Multi-hop	هل تسقط عقوبة الغش عن الطالب إذا تم اكتشاف غشه في اليوم التالي للامتحان؟	Passed
Multi-hop	حدثت مشاجرة كبيرة أدت إلى إخلال بالنظام، هل يشترط تشكيل لجنة تحقيق لإيقاع العقوبة؟	Passed

2.9 Validation and Findings

During live inference cycles, the ParentDocumentRetriever successfully matched short conversational child chunks and dynamically pulled the full, multi-paragraph

text. This dual-layer architecture achieved zero context truncation and delivered complete factual accuracy, validating the core design requirements of the project.